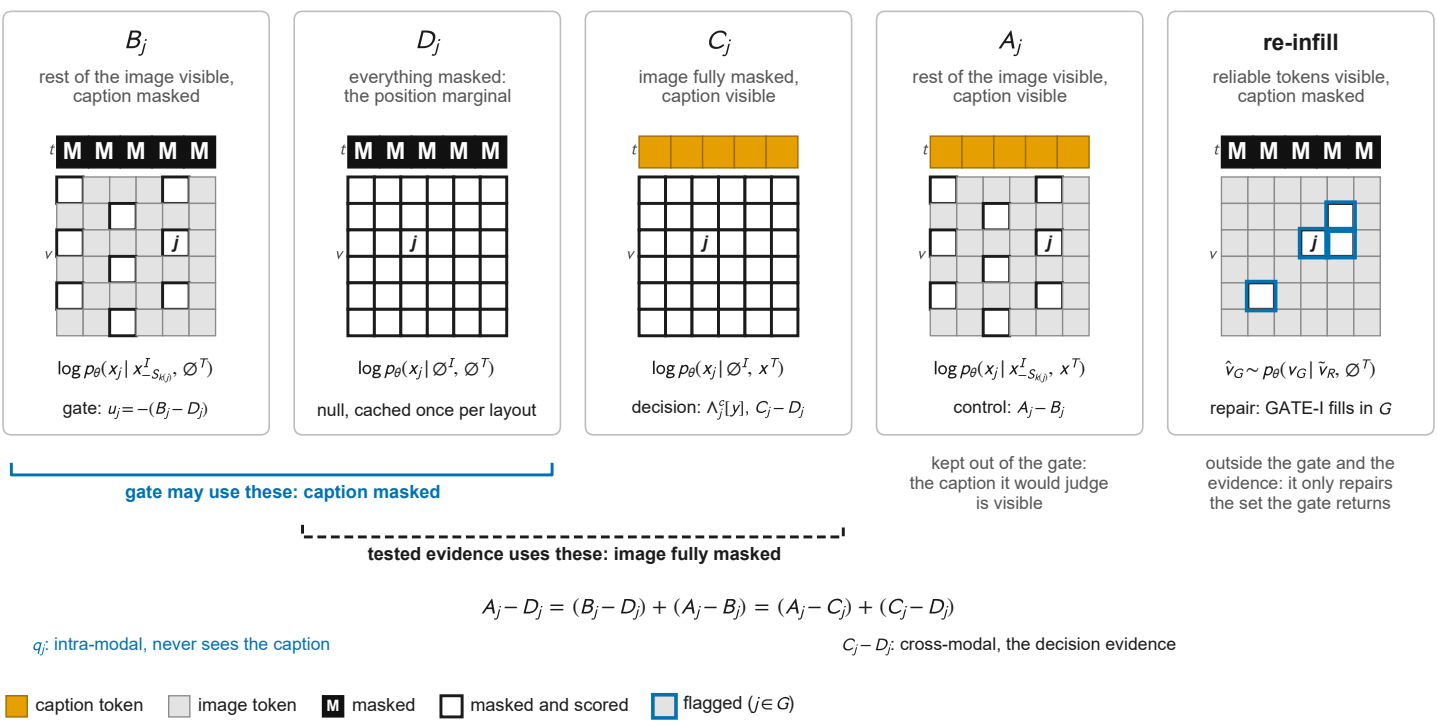

B_j
rest of the image visible, caption masked
t
M M M M M
v
j
$\log p_\theta(x_j \mid x^I_{\setminus S_{k(j)}}, \varnothing^T)$
gate: $u_j = -(B_j - D_j)$
D_j
everything masked: the position marginal
t
M M M M M
v
j
$\log p_\theta(x_j \mid \varnothing^I, \varnothing^T)$
null, cached once per layout
C_j
image fully masked, caption visible
t
v
j
$\log p_\theta(x_j \mid \varnothing^I, x^T)$
decision: $\Lambda^c_j[y]$, $C_j - D_j$
A_j
rest of the image visible, caption visible
t
v
j
$\log p_\theta(x_j \mid x^I_{\setminus S_{k(j)}}, x^T)$
control: $A_j - B_j$
re-infill
reliable tokens visible, caption masked
t
M M M M M
v
j
$\hat{v}_G \sim p_\theta(v_G \mid \tilde{v}_R, \varnothing^T)$
repair: GATE-I fills in G
gate may use these: caption masked
kept out of the gate: the caption it would judge is visible
outside the gate and the evidence: it only repairs the set the gate returns
tested evidence uses these: image fully masked
$A_j - D_j = (B_j - D_j) + (A_j - B_j) = (A_j - C_j) + (C_j - D_j)$
q_j: intra-modal, never sees the caption
$C_j - D_j$: cross-modal, the decision evidence
caption token
image token
M masked
masked and scored
flagged ($j \in G$)